

Andrew William Baggaley

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EMPLOYMENT

Sept 2025–onwards	Prof in mathematics and AI, Lancaster University, UK
2024–2025	Reader in applied mathematics, Newcastle University, UK
2022–2024	Senior Lecturer of mathematics, Newcastle University, UK
2015–2022	Lecturer of mathematics, Newcastle University, UK
2012–2015	Lecturer of mathematics, University of Glasgow, UK
2010–2012	Post-doctoral research associate, Newcastle University, UK
	Mathematical modelling of the Neolithic of Europe
	PDRA Supervisor: Dr. Graeme R. Sarson

EDUCATION

2006–2009	Ph.D. Applied Mathematics Supervisors: Prof. Anvar Shukurov and Prof. Carlo F. Barenghi Newcastle University, UK
2002–2006	MMath. First class with Hons. Newcastle University, UK

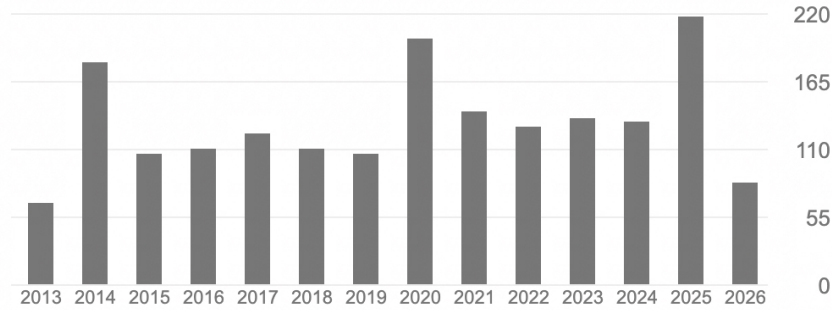
GRANTS AND AWARDS

2024	EPSRC “Statistical models of the decay of 2D quantum turbulence” (PI)	£100K
2024	DEFRA “Developing a prototype landscape incursion risk emulator” (Co-I)	£130K
2022	STFC “Superfluid dynamics in neutron stars” (Co-I)	£280K
2022	NERC “Discipline Hopping for Environmental Solutions” (Co-I)	£45K
2021	Leverhulme Trust “Turbulence in Quantum Ferrofluid” (PI)	£160K
2020	EPSRC “Modelling and inference of UK tree pandemics” (PI)	£200K
2014	ESF exploratory workshop “Reconnection events in classical, quantum and magnetized fluids” (PI)	£12K

IMPACT

74+ refereed journal articles (incl. 3 PRL, 3 PNAS)
 $N \approx 1650$ citations, $h = 25$, $i_{10} = 38$ (Google)
Section Head & Director of UG Teaching, School of Mathematical Sciences, Lancaster
Director of Applied Mathematics, School of Mathematics, Statistics & Physics, Newcastle
(Co-)organiser or member of LOC of 6 major international conferences
Member of the organising committee of the UK Quantum Fluids Network & Webinar Series:
<https://uk-quantum-fluids-network.github.io/webinars/>
Member of Editorial Board, PLOS One
Peer-reviewer for: PNAS, Nature, Nature Physics, PRL, MNRAS, J. R. Soc. Interface, ...

Citations per year



TEACHING AND SUPERVISION

Lecture courses, student questionnaire scores are available upon request.

Undergraduate Modelling and Programming (~ 100 students), Lancaster	2026–
Undergraduate Hydrodynamic Instabilities (~ 30 students), Newcastle	2023–2025
Undergraduate Group project, module leader (~ 120 students), Newcastle	2022–2025
Undergraduate PDE course (22 lectures, ~ 100 students), Newcastle	2021–23
Undergraduate ODE & PDE course (22 lectures, ~ 120 students), Newcastle	2017–21
Undergraduate Group project, module leader (~ 160 students), Newcastle	2016–2020
Undergraduate Problem solving course (~ 160 students), Newcastle	2016–
Undergraduate PDE course (22 lectures, ~ 100 students), Newcastle	2015
Undergraduate ODE & PDE course (30 lectures, ~ 100 students), Glasgow	2012–2014
Undergraduate Calculus course (22 lectures, ~ 100 students), Glasgow	2013

Undergraduate project supervision - Supervised 50+ undergraduate (group and individual) projects on a range of applied maths problems including dynamo theory, coherent animal motion, archaeology and fluid dynamics.

External roles, External examiner, Aston University Mathematics BSc, 2023–

Post Doctoral Research Associates

Nick Keepfer	<i>Reinforcement Learning in atomic condensates</i>	2025-2026
Gary Liu	<i>Superfluid dynamics in Neutron stars</i>	2023-2026
Srivatsa Badariprasad	<i>Turbulence in dipolar condensates</i>	2021-2024
Ryan Doran	<i>Vortices in two-component BECs</i>	2021-2023
Laura Wadkin	<i>Modelling and inference of UK tree pandemics</i>	2021-2022
Hannah Kreczak	<i>Biofouling of microplastics</i>	2020-2022
Sirio Orozco-Fuentes	<i>Forecasting of plant epidemics</i>	2016-2018

Postgraduate students

Matthew Richardson	<i>Technological applications of BECs</i>	Newcastle PhD 2024-
Axa-Maria Laaperi	<i>Mathematical and statistical models of wildfires</i>	Newcastle PhD 2023-
Julie Thomas	<i>Superfluid dynamics in Neutron stars</i>	Newcastle PhD 2022-
Peter Stasiak	<i>Finite temperature quantum turbulence</i>	Newcastle PhD 2022-
Matt Dopson	<i>Network modelling of food webs</i>	Newcastle PhD 2022-
Jamie McKeown	<i>Resilient Treescapes: a mathematical approach</i>	Newcastle PhD 2022-
Richard Tattersall	<i>Vortex Dynamics in Quantum Fluids</i>	Newcastle PhD 2021-2025
George Grimes	<i>Quantised vortex dynamics</i>	Newcastle MPhil 2019-2022
John Holden	<i>Global models of forest epiphytotics</i>	Leeds/Newcastle PhD 2017-2022
Sultan Aylobi	<i>Network models of forest epiphytotics</i>	Newcastle PhD 2017-2021
Jack Walton	<i>Bayesian inference of cooperative motion</i>	Newcastle PhD 2016-2021
Hayley Moore	<i>Constraining models of collective animal motion</i>	Newcastle PhD 2016-2021
Em Rickinson	<i>Quantum turbulence</i>	Newcastle PhD 2015-2020
Scott Richardson	<i>Modelling of Active Suspensions</i>	Glasgow PhD 2012-2017
Lucy Sherwin-Robson	<i>Quantum turbulence</i>	Newcastle PhD 2011-16
Elena Ardito	<i>2D Turbulence in Bose-Einstein condensates</i>	Glasgow MSc 2013

Postgraduate examinations

Anirudh Sivakuma	Bharathidasan University, India	PhD 2026
Thomas Newman	Lancaster (internal)	PhD 2025
Thomas Flyn	Newcastle (internal)	PhD 2024
Kate Brown	Newcastle (internal)	PhD 2024
Henrik Carlson	Olso	PhD 2023
Holly Middleton-Spencer	Newcastle (internal)	PhD 2023
Kieran Peel	Newcastle (internal)	PhD 2022
Nick Keepfer	Newcastle (internal)	PhD 2022
Ryan Doran	Newcastle (internal)	PhD 2021
Laura Wadkin	Newcastle (internal)	PhD 2021
Cetin Evirgen	Newcastle (internal)	PhD 2020
Adam Griffin	Warwick	PhD 2020
Sarah Jowett	Newcastle (internal)	MPhil 2018
Niklas Hietla	Aalto	PhD 2017
Alex Williamsi	Manchester	MPhil 2015
Tianhui Zhui	Manchester	MPhil 2015
Umar Qureshi	Glasgow (internal)	PhD 2014
Edward Ashley Guise	Lancaster	PhD 2014

REFERENCES

Prof. Nick Parker

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Prof. Paul Bushby

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Dr Jason Laurie

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Andrew William Baggaley

Refereed Journal Articles

- [74] DOPSON, M, PARKER N.P., WADKIN, L., BAGGALEY, A.W., PAKEMAN, R.J., SMITH, S.W. & EVANS, D.M. The impacts of grazing pressure by large herbivores on short-tailed field vole (*Microtus agrestis*) population cycles: results from a long-term upland experiment, in submission 2026.
- [73] PRASAD, S. B., PARKER, N. G., & BAGGALEY, A. W. Crow instability of vortex lines in dipolar superfluids. *Scientific Reports*, **15**, 33364 (2025).
- [72] BAINS, J. S., BAGGALEY, A. W., & CROZE, O. A. Drift velocity of bacterial chemotaxis in dynamic chemical environments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* (2025).
- [71] QUICK, S. R., BAINS, J. S., GERDT, C., WALKER, B., GOLDSTONE, E. B., JAKUSZEIT, T., ET AL. A close unicellular animal relative and predator of schistosomes exhibits chemokinesis in response to proteins and peptides from its prey. *PLoS Pathogens*, **21**, e1013440 (2025).
- [70] TATTERSALL, R. J., BAGGALEY, A. W., & BILLAM, T. P. Out-of-equilibrium behavior of quantum vortices: A comparison of point vortex dynamics and Fokker–Planck evolution. *Physical Review A*, **112**, 013313 (2025).
- [69] STASIAK, P. Z., XING, Y., ALIHOSSEINI, Y., BARENGHI, C. F., BAGGALEY, A., GUO, W., ET AL. Experimental and theoretical evidence of universality in superfluid vortex reconnections. *Proceedings of the National Academy of Sciences*, **122**, e2426064122 (2025).
- [68] LIU, I. K., BAGGALEY, A. W., BARENGHI, C. F., & WOOD, T. S. Vortex avalanches and collective motion in neutron stars. *The Astrophysical Journal*, **984**, 83 (2025).
- [67] PRASAD, S. B., PARKER, N. G., & BAGGALEY, A. W. Vortex-pair dynamics in three-dimensional homogeneous dipolar superfluids. *Physical Review A*, **109**, 063323 (2024).
- [66] DOYLE, M. J., GOLOV, A. I., WALMSLEY, P. M., & BAGGALEY, A. W. Modelling Turbulent Flow of Superfluid 4 He Past a Rough Solid Wall in the $T = 0$ Limit. *Journal of Low Temperature Physics*, 1-13, (2024).
- [65] TATTERSALL, R. J., BAGGALEY, A. W., & BILLAM, T. P. Non-equilibrium Dynamics of Vortices in Two-Dimensional Quantum Gases: Determining the Dynamical Scaling Region Using the Mahalanobis Distance. *Journal of Low Temperature Physics*, 1-14, (2024).
- [64] LIU, I. K., PRASAD, S. B., BAGGALEY, A. W., BARENGHI, C. F., & WOOD, T. S. Vortex depinning in a two-dimensional superfluid. *Journal of Low Temperature Physics*, 1-21, (2024).
- [63] WADKIN, L. E., HOLDEN, J., ETTELAIE, R., HOLMES, M. J., SMITH, J., GOLIGHTLY, A., & BAGGALEY, A. W. Estimating the reproduction number, R_0 , from individual-based models of tree disease spread. *Ecological Modelling*, **489**, 110630, (2024)
- [62] DORAN, R., BAGGALEY, A. W., & PARKER, N. G. Vortex solutions in a binary immiscible Bose-Einstein condensate. *Physical Review A*, **109**, 023318, (2024).
- [61] STASIAK, P. Z., BAGGALEY, A. W., KRSTULOVIC, G., BARENGHI, C. F., & GALANTUCCI, L. Cross-Component Energy Transfer in Superfluid Helium-4. *Journal of Low Temperature Physics*, 1-12, (2024).
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- [54] G.S.E. GRIMES AND A. W. BAGGALEY, Approach and separation of bundles of quantized vorticity, *Phys. Rev. Fluids*, **7**, 034701, (2022).
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- [41] S. OROZCO-FUENTES, G. GRIFFITHS, M. J. HOLMES, R. ETTELAIE, J. SMITH, A. W. BAGGALEY, & N. G. PARKER Early warning signals in plant disease outbreaks, *Ecological Modelling* **393**, 12-9, (2019).
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- [14] BAGGALEY, A.W., SHERWIN, L.K., BARENGHI, C.F. & SERGEEV, Y.A. Thermally- and mechanically-driven quantum turbulence in helium II, *Physical Review B* **86**, 104501 (2012).
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- [7] BAGGALEY, A.W. , BARENGHI, C.F. Tree method for quantum vortex dynamics *Journal of Low Temperature Physics* **166**, 3–20 (2012).
- [6] BAGGALEY, A.W. & BARENGHI, C.F. Quantum turbulent velocity statistics and quasi-classical limit *Physical Review E* **84**, 067301 (2012).
- [5] BAGGALEY, A.W. , BARENGHI, C.F. Vortex-density fluctuations in quantum turbulence *Physical Review B* **84**, 020504 (2011).
- [4] BAGGALEY, A.W. , BARENGHI, C.F. Spectrum of turbulent Kelvin-waves cascade in superfluid helium *Physical Review B* **83**, 134509 (2011).
- [3] BAGGALEY, A.W. , SHUKUROV, A. , BARENGHI, C.F. & SUBRAMANIAN, K. Fluctuation dynamo based on magnetic reconnections *Astronomische Nachrichten* **331**, 46–62 (2010).
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